

RESEARCH ARTICLE

Compressive Signal Processing for Estimating Range-Velocity-AoA in FMCW Radar Applications

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ABSTRACT Frequency-modulated continuous wave (FMCW) radars are known to accurately estimate the parameters of targets with low-cost and low-power transceiver systems. This work shows that some features associated with the spectrum of estimated parameters allow one to integrate the compressive sampling (CS) theory into the FMCW signal processing. To this end, we establish a new analytical framework through a tensor format to facilitate a systematic and convenient FMCW signal processing model. By observing the sparsity feature in the tensor of FMCW spectrum, we develop an extensive theoretical analysis to justify the use of CS theory in the FMCW radars, and it enabled us to propose a novel scheme, namely compressive FMCW signal processing, for estimating range, velocity, and angle-of-arrival (AoA) of the targets. The addition of CS theory allows our proposed scheme to use sampling-rates below Nyquist criterion, thus minimizing the number of sampled data and mitigating the issues related to high-rate and power-hungry analog-to-digital converter (ADC) in extremely high-frequency radar applications. Furthermore, the proposed compressive FMCW signal processing also significantly reduces the number of radio-frequency (RF) front-end elements necessary for estimating AoA, leading to a further saving in cost and power consumption of the FMCW radars. Despite having sub-Nyquist sampling-rates and reduced RF front-end elements, the performance evaluations show that our CS-based approach maintains the estimation capability and accuracy of conventional FMCW signal processing.

INDEX TERMS FMCW radars, compressive sampling, range, velocity, angle-of-arrival, parameters estimation.

I. INTRODUCTION

The frequency-modulated continuous wave (FMCW) radars have become popular tools to detect targets and estimate their parameters due to several key benefits, such as low-complexity radar transceiver [1], low-power requirement for signal transmission [2], and highly accurate measurement of parameters [3]. Typically, the estimated parameters are range and velocity of the targets from the radar. The features of FMCW radar make it attractive for various applications such as LiDAR system [4], automotive radars [5], synthetic aperture radar (SAR) [6], weather radar [7], through-the-wall radar [8], and maritime radar [9].

FMCW radar transmits a continuous sinusoid wave signal called chirp, whose frequency linearly increases over time, and the receiver intercepts the reflected chirp signal from the targets. Since the reflected chirp signal is a time-delayed version of the transmitted one, a frequency difference (or beat frequency) between those two chirp signals is exploited to estimate the targets' parameters. The beat frequency is conveniently measured by mixing the two chirp signals and passing the mixer output through a low-pass filter. It is shown in [3] that the beat frequency is proportional to the distance of targets from the receiving antenna, thus enabling range measurement of the targets. In addition, the velocity of moving targets can also be estimated by transmitting a signal containing several equi-spaced chirps with a linear sawtooth frequency modulation [3], [10], [11]. These chirps have a specific duration, and they are sent in one FMCW

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